

Standard Operating Procedure for Huawei PE Router

- ALARM_5646

Node: Huawei-PE-Delhi-NCR-11 | **Vendor:** Huawei | **Severity:** Critical | **Alarm Code:** ALARM_5646

Description

Huawei Provider Edge Router (PE) - Delhi NCR Node (Node ID: Huawei-PE-Delhi-NCR-11) - Critical Alarm: ALARM_5646

Potential Root Causes

Cause	Probability
Hardware failure in the routing engine	High
Software bug in the routing protocol	Medium
Misconfiguration of BGP peering	Low

Diagnostic Commands

```
show system internal routing-engine statistics
show system internal routing-engine processcpu
show bgp summary
show bgp neighbors
```

Resolution Steps

- Step 1: {'Step': 'Step 1: Check routing-engine statistics', 'Action': "Run the command 'show system internal routing-engine statistics' to check for hardware failure", 'CLI_Command': 'show system internal routing-engine statistics'}
- Step 2: {'Step': 'Step 2: Check routing-engine process CPU utilization', 'Action': "Run the command 'show system internal routing-engine processcpu' to check for software bug", 'CLI_Command': 'show system internal routing-engine processcpu'}
- Step 3: {'Step': 'Step 3: Check BGP peering configuration', 'Action': "Run the command 'show bgp summary' and 'show bgp neighbors' to check for misconfiguration", 'CLI_Command': 'show bgp summary; show bgp neighbors'}
- Step 4: {'Step': 'Step 4: Restart routing engine if necessary', 'Action': "Run the command 'restart routing-engine' to restart the routing engine if necessary", 'CLI_Command': 'restart routing-engine'}

Post-Check Verification

```
show system internal routing-engine statistics
```

```
show system internal routing-engine processcpu  
show bgp summary  
show bgp neighbors
```